



THE PROBLEM

1.5 billion people at risk of chronic liver disease. Biopsy is invasive. Commercial elastography systems cost £30K–£80K and are closed, inflexible single-point devices. No open-source alternative exists with validated physics for clinical decision-making.

£50

BOM cost per board

£50K+

Commercial competitor price

8

Channels, spatial mapping

\$680M

SAM: portable elastography

THE SOLUTION

TurboQuant V5 is an open-hardware, 8-channel ultrasound acquisition platform integrating $\pm 100V$ HV pulser, T/R switching, analog MUX, and low-noise amplifiers on a single 100x80 mm PCB. Built on the Red Pitaya SDR platform. Total BOM under £50. Peer-review-ready Bayesian framework for viscoelastic parameter estimation — the first open system with rigorously validated physics for clinical elastography.

TRACTION & MILESTONES

Completed:

- 5 hardware iterations (V1→V5), schematic-complete, ERC clean
- Published-ready Bayesian MCMC framework for viscoelastic inversion
- 2D FDTD simulation + dispersion validation (30% → <2% error)
- Component selection validated: DG408, IRF830, OPA1641, 74HCT595
- 15+ open-source contributors on GitHub

Next 6 months:

- 20-unit prototype batch + bench validation
- Phantom studies (ATS 539 liver phantom)

MARKET & REVENUE MODEL

Segments:

- Research kits: £299 (BOM £50) — universities, NDE contractors
- Clinical tier: £899 + £199/year service — early adopters
- SaaS/API: £99/year — advanced DSP, cloud analytics, ML models

Market sizing (Year 3):

- TAM: \$4.2B | SAM: \$680M | SOM: \$85M
- Target: 2,000 units + 500 service contracts
- Revenue target: £1.2M, 45% blended margin

COMPETITIVE POSITION

FibroScan (£30K+, closed, single-point) | Siemens ACUSON (£50K+, closed ecosystem) | Supersonic Aixplorer (£80K+, research-only)
TurboQuant: the only platform competing on cost (£50 BOM), openness (GitHub + open API), AND scientific rigor (validated Bayesian physics).

THE TEAM

Founder/Lead Engineer: 5 iterations V1→V5, FPGA, ultrasound hardware, embedded systems
Research: Bayesian inverse problem specialist — MCMC, FDTD, phantom physics
Advisors: NHS hepatologist (regulatory pathway) + MedTech entrepreneur (SaaS)

THE ASK

£75,000

20% equity | 6-month runway

→ 20 prototypes → phantom validation → journal publication → SaaS launch